

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>
we know john drove for 3 hours at the start at 60 miles per hour so we can do 3 times 60 to work out how far he got in the first part
so we do 3 times 60 which equals 180 so now we know he's 180 miles away from home
then the question says he spends 2 hours in stand still traffic so we know that doesn't add any distance so we can ignore that part
then it says he drives for half a hour at 30 miles per hour so we do half of 1 hour which is 0 point 5 and then do 0 point 5 times 30 to work out the distance
so 0 point 5 times 30 equals 15 so now we know he's done 15 miles in that half hour
so now we take that 15 off the 180 so 180 take away 15 equals 165 so now we know he's 165 miles from home
then the question says in the next part he drives for the rest of the 4 hours at 80 miles an hour we know the total time is 4 hours and we know he's already spent 2 point 5 hours so we take 4 take away 2 point 5 which equals 1 point 5
so now we know he drives for 1 point 5 hours at 80 miles an hour so we do 1 point 5 times 80
1 point 5 times 80 equals 120
so now we know he's done another 120 miles
so now we take the 120 off the 165 so 165 take away 120 equals 45
so now we know he's 45 miles away from home
</think><answer>45</answer>